

Consequently, the only possible measurement outcomes are 00 and 11. Since qubit 0 has an equal probability of being measured as 0 or 1, we obtain approximately equal numbers of 00 and 11 outcomes over 1000 shots. The small variations in the counts across the four executions are simply a consequence of the probabilistic nature of quantum measurement.

Let us now slightly modify the program *qiskit9.py* to see what happens when the two qubits are not entangled. The program *qiskit9_mod1.py* is obtained by replacing Line 5 with *qc.h(1)*. Here, instead of using a CNOT gate to entangle the two qubits, we apply a Hadamard gate independently to qubit 1 as well. Figure 6 shows the results of four successive executions of *qiskit9_mod1.py*. As we can see from the figure, all four possible combinations—00, 01, 10, and 11—occur with approximately equal probability, with roughly 250 occurrences of each in 1000 shots. This is exactly what we would expect when the two qubits are independent. So, for this circuit, the moral of the story is simple: no CNOT, no entanglement!

Now, let us do some myth busting. Is it possible to communicate faster than light using entangled particles? Let us consider a theoretical scenario. You and your friend share a collection of entangled qubit pairs and then travel light years apart. You measure your qubit and suppose you obtain 1. If your friend measures the other qubit in the same computational basis, he will also obtain 1 with certainty. It may therefore appear that you have somehow

```
(qiskit_env) # python qiskit9_mod1.py
{'01': 268, '11': 237, '10': 242, '00': 253}
(qiskit_env) # python qiskit9_mod1.py
{'11': 252, '01': 251, '10': 246, '00': 251}
(qiskit_env) # python qiskit9_mod1.py
{'11': 254, '10': 253, '00': 250, '01': 243}
(qiskit_env) # python qiskit9_mod1.py
{'11': 274, '01': 235, '00': 250, '10': 241}
```

Figure 6: Output of the program *qiskit9_mod1.py*

communicated the result of your measurement instantaneously. But have you actually sent any information faster than light? The answer is a big ‘No’.

Suppose you want to send a single bit of information—say, 0—to your friend. You cannot force your entangled qubit to produce 0 when measured. The outcome is fundamentally random. You could keep measuring identically prepared entangled pairs until you happen to obtain 0, but your friend has no way of knowing which pair produced that result. Suppose, for example, that you share five entangled pairs. You measure the first two pairs and obtain 1, while the third measurement gives you the 0 you wanted to communicate. Here comes the catch: how will you tell your friend that the third pair is the one he should examine? You must send that information through a classical communication channel, and classical information cannot travel faster than light. Your friend can therefore discover the correlation only after receiving information through an ordinary communication channel. Thus, although

entanglement produces correlations that cannot be explained by classical physics, it cannot be used for faster-than-light communication. Faster-than-light communication

through entanglement is therefore a science-fiction plot device, not a feature of quantum mechanics.

Now, it is time to conclude our discussion. In this article, we saw how a quantum computing simulator can be configured to imitate some of the noisy behaviour of a real quantum processor. We then took a step further by executing a quantum program on an actual superconducting quantum computer through the IBM Quantum Platform. Finally, we shifted our gears from the world of single qubits to the exciting realm of multi-qubit systems, where we encountered the CNOT gate and the fascinating phenomenon of quantum entanglement. These concepts open the door to the truly quantum world of computing, where the real power of quantum systems begins to unfold. In the next article in this series, we will delve deeper into entanglement, explore Bell states and related concepts, and take a closer look at the various sources of errors that affect today’s superconducting quantum computers. **END** 

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